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Course: 21CSC202J - Operating Systems

Unit – 3 : Session – 15 : SLO – 1 | Banker's Algorithm — Safety and Resource Request

Need = Max - Allocation

```
         R0 R1 R2
P0       3  1  0
P1       1  2  0
P2       0  2  1
```

`Available = (2 2 0)`

(a) The state is valid (safe)

P1: need (1 2 0) <= (2 2 0) -> finishes, Available = (2 5 1)

P2: need (0 2 1) <= (2 5 1) -> finishes, Available = (3 5 3)

P0: need (3 1 0) <= (3 5 3) -> finishes, Available = (4 5 5)

Safe sequence <P1, P2, P0>; total resources (4 5 5) cover every Max, so the state can occur.

(b) P0 requests one unit of R1

Request (0 1 0) <= Need P0 (3 1 0) and <= Available (2 2 0) -> pretend to grant

New Available = (2 1 0), P0 Allocation = (1 1 2), P0 Need = (3 0 0)

P0 needs (3 0 0) > (2 1 0); P1 needs (1 2 0) > (2 1 0); P2 needs (0 2 1) > (2 1 0)

No process can finish -> the resulting state is UNSAFE.

The system denies the request; P0 must wait and the old state is restored.